

EXAMEN LA ANALIZA MATEMATICA II

I. Fie

$$A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \geq 1\}$$

si fie B multimea din $\mathbb{R}^2$ marginita de dreptele $x = 2$, $y = -x$ si $y = \sqrt{3}x$. Daca $D = A \cap B$, sa se calculeze

$$\iint_D x^3 dx dy.$$

Nota: Nu trebuie sa verificati ca multimea D este masurabila Jordan.

II. Sa se arate ca ecuatia

$$x^2 + y^2 + z^2 + 3xy - 5xz - 5yz + 4 = 0$$

defineste implicit functia $z = z(x, y)$ intr-o vecinatate a lui $(1, 1, 1)$ si sa se calculeze $\frac{\partial z}{\partial x}(1, 1)$, $\frac{\partial z}{\partial y}(1, 1)$, $\frac{\partial^2 z}{\partial x^2}(1, 1)$, $\frac{\partial^2 z}{\partial y^2}(1, 1)$ si $\frac{\partial^2 z}{\partial y \partial x}(1, 1)$.

Sa se determine daca $(1, 1)$ este punct de extrem local pentru functia implicita $z = z(x, y)$ de mai sus si in caz afirmativ sa se determine natura lui (adica daca este de maxim sau de minim)

III. Fie A multimea din $\mathbb{R}^3$ marginita de planele $z = 0$, $y = 0$, $x + 2y + 4z = 8$, $x = 1$ si $x = 2$ si

$$B = A \setminus \left\{ (x, y, z) \in \mathbb{R}^3 \mid y \leq 1, x \in \left\{ 1 + \frac{1}{n^2} \mid n \in \mathbb{N}^* \right\} \right\}.$$

1) Aratati ca multimea A este masurabila Jordan

2) Calculati

$$\iiint_A x^2 dx dy dz$$

3) Decideti daca multimea B este masurabila Jordan si in caz afirmativ calculati

$$\iiint_B x^2 dx dy dz.$$

IV. Fie $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = 5x^2 + 3xy + y^2.$$

Sa se determine extreme globale ale functiei f pe multimea $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ (adica determinati $\sup_D f(x, y)$ si $\inf_D f(x, y)$).

Nota. Timpul de lucru este de 2 ore. Fiecare subiect se noteaza cu note de la 1 la 10. Nota obtinuta la aceasta lucrare este media aritmetica a celor 4 note.

Rezolvarile trebuie scanate si trimise impreuna cu lista de subiecte sub forma unui **singur** fisier pdf la adresele radu-bogdan.munteanu@g.unibuc.ro si radu.munteanu@unibuc.ro.

Examen Analiză II

$$1. A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \geq 1\}$$

B mulțimea din $\mathbb{R}^2$ mărginită de dreptele $x=2$, $y=-x$, $y=\sqrt{3}x$

$$\Delta = A \cap B, \iint_{\Delta} x^3 dx dy$$

$$\begin{cases} x=2 \\ y=\sqrt{3}x \end{cases} \rightarrow \begin{cases} x=2 \\ y=2\sqrt{3} \end{cases}$$

$$\begin{cases} y=-x \\ x=2 \end{cases} \rightarrow \begin{cases} y=-2 \\ x=2 \end{cases}$$

$$B = B_1 \cup B_2$$

$$B_1 = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 2, 0 \leq y \leq \sqrt{3}x\}$$

$$B_2 = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 2, -x \leq y \leq 0\}$$

$$\iint_{A \cap B} x^3 dx dy = \iint_B x^3 dx dy - \iint_{\Delta} x^3 dx dy$$

$A \cap B$

Unde Δ este regiunea albastră

$B \rightarrow$ triunghi

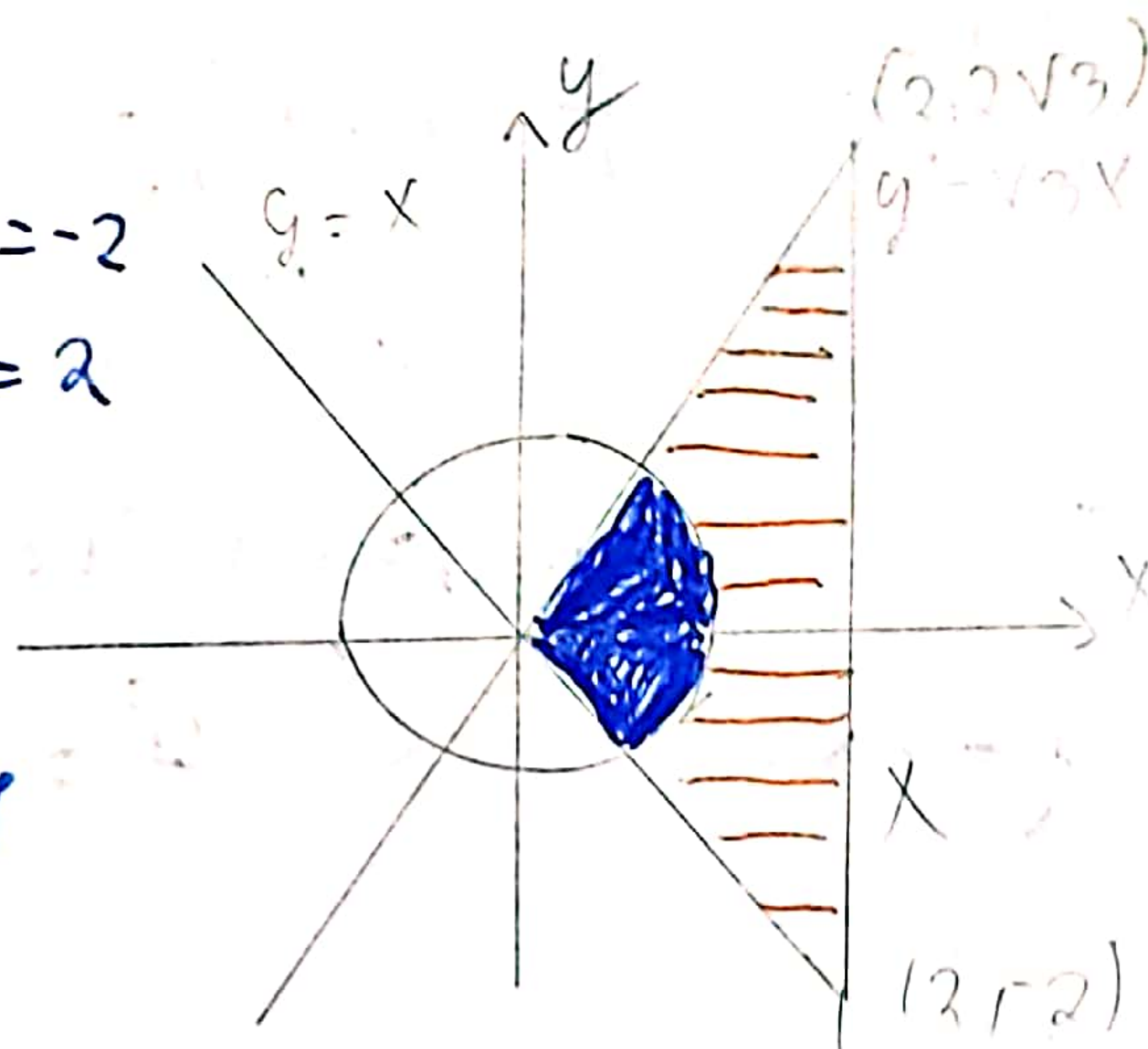
$$B = B_1 \cup B_2, B_1 = \{(x, y) \mid y \in [-2, 0], -y \leq x \leq 2\}$$

$$B_2 = \{(x, y) \mid y \in [0, 2\sqrt{3}], \frac{y}{\sqrt{3}} \leq x \leq 2\}$$

$$\Rightarrow \iint_B x^3 dx dy = \iint_{B_1} x^3 dx dy + \iint_{B_2} x^3 dx dy = I_1 + I_2$$

$$I_1 = \int_{-2}^0 \int_{-y}^2 x^3 dx dy = \int_{-2}^0 \left[\frac{x^4}{4} \right]_{-y}^2 dy = \int_{-2}^0 \left(\frac{2^4}{4} - \frac{y^4}{4} \right) dy$$

$$= \int_{-2}^0 \left(4 - \frac{y^4}{4} \right) dy = \left[4y - \frac{y^5}{20} \right]_{-2}^0 = \frac{32}{5}$$



$$I_2 = \int_0^{2\sqrt{3}} \int_{\frac{y}{\sqrt{3}}}^2 x^3 dx dy = \int_0^{2\sqrt{3}} \left[\frac{x^4}{4} \right]_{\frac{y}{\sqrt{3}}}^2 dy = \int_0^{2\sqrt{3}} \frac{16}{4} - \frac{y^4}{4} dy$$

$$= \int_0^{2\sqrt{3}} \left(4 - \frac{y^4}{36} \right) dy = 4y - \frac{y^5}{180} \Big|_0^{2\sqrt{3}} = \frac{32\sqrt{3}}{5}$$

Continuarea este pe pagina 7.

π

$$x^2 + y^2 + z^2 + 3xy - 5xz - 5yz + 4 = 0, z = z(x, y) \text{ într-o vec. a lui } (1, 1, 1)$$

$$x \cdot \frac{\partial z}{\partial x}(1, 1), \frac{\partial z}{\partial y}(1, 1), \frac{\partial^2 z}{\partial x^2}(1, 1), \frac{\partial^2 z}{\partial y^2}(1, 1), x \cdot \frac{\partial^2 z}{\partial y \partial x}(1, 1)$$

$(1, 1)$ punct de extrem local și natura sa, în caz afirmativ

$$F: \mathbb{R}^3 \rightarrow \mathbb{R}, F(x, y, z) = x^2 + y^2 + z^2 + 3xy - 5xz - 5yz + 4$$

$$\frac{\partial F}{\partial x} = 2x + 3y - 5z \quad \frac{\partial F}{\partial y} = 2y + 3x - 5z \quad \frac{\partial F}{\partial z} = 2z - 5x - 5y$$

i) F de clas C^1

$$ii) F(1, 1, 1) = 1 + 1 + 1 + 3 - 5 - 5 + 4 = 0$$

$$iii) \frac{\partial F}{\partial z}(1, 1, 1) = -8 \neq 0$$

Atunci există o vecinătate deschisă V_0 a lui $V(1, 1)$, există V_0 vecinătate deschisă a lui $V(1, 1)$ și o unică funcție $z = z(x, y), z: V_0 \rightarrow V_0$.

$$1) z(1, 1) = 1$$

$$2) F(x, y, z(x, y)) = 0, \forall (x, y) \in V$$

3) z de clasă C^2

$$\frac{\partial z}{\partial x}(1, 1) = \frac{\frac{\partial F}{\partial x}(1, 1, z(1, 1))}{\frac{\partial F}{\partial z}(1, 1, z(1, 1))} = \frac{2 + 3 - 5z(1, 1)}{-5 - 5 + 2z(1, 1)} \stackrel{z(1, 1)=1}{=} \frac{0}{-12} = 0$$

$$\frac{\partial z}{\partial y}(1, 1) = \frac{2 + 3 - 5z(1, 1)}{-5 - 5 + 2z(1, 1)} = 0$$

$$\frac{\partial^2 z}{\partial x^2}(1,1) = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{(2 - 5 \frac{2z}{2x}) (5x + 5y - 2z) - (2x + 3y - 5z) (5 - 2 \frac{2z}{2x})}{(5x + 5y - 2z)^2}$$

$$= \frac{(2 - 5 \cdot 0) (5 + 5 - 2) - (2 + 3 - 5) \cdot (5 - 2 \cdot 0)}{(5 + 5 - 2)^2} = \frac{1}{4} \quad \frac{\partial^2 z}{\partial y^2} = \frac{1}{4}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{(3 - 5 \frac{2z}{2y}) (5x + 5y - 2z) - (2x + 3y - 5z) (5 - 2 \frac{2z}{2y})}{(5x + 5y - 2z)^2}$$

$$= \frac{3}{8}$$

$$\begin{cases} F(x, y, z) = 0 \\ \frac{\partial F}{\partial z}(x, y, z) \neq 0 \\ \frac{\partial F}{\partial x}(x, y, z) = 0 \\ \frac{\partial F}{\partial y}(x, y, z) = 0 \end{cases} \Leftrightarrow \begin{cases} x^2 + y^2 + z^2 + 3xy - 5xz - 5yz + 4 = 0 \\ 2x - 5x - 5y \neq 0 \\ 2x + 3y - 5z = 0 \\ 2y + 3x - 5z = 0 \end{cases}$$

una dintre solutii este: $(1, 1, 1)$

$z = z(x, y)$ cu punctul critic $(1, 1)$ a. i. $z(1, 1) = 1$

$$M_x(x, y) = \begin{pmatrix} \frac{\partial^2 x}{\partial x^2} & \frac{\partial^2 z}{\partial x \partial y} \\ \frac{\partial^2 z}{\partial y \partial x} & \frac{\partial^2 z}{\partial y^2} \end{pmatrix} \Rightarrow M_z(1, 1) = \begin{pmatrix} \frac{1}{4} & \frac{3}{8} \\ \frac{3}{8} & \frac{1}{4} \end{pmatrix}$$

$$\begin{vmatrix} \Delta_1 = \frac{1}{4} > 0 \\ \Delta_2 = -\frac{5}{64} < 0 \end{vmatrix} \Rightarrow (1, 1) \text{ nu este punct de extrem pentru } \text{funcția } z.$$

III

A din $\mathbb{R}^3$ mărginită de planele $z=0, y=0, x+2y+4z=8, x=1, x=2$

$$\alpha: B = A \cap \{(x, y, z) \in \mathbb{R}^3 \mid y \leq 1, x \in [1 + \frac{1}{n^2}, 1 + \frac{1}{(n+1)^2}] \mid n \in \mathbb{N}^*\}$$

1) A este măsurabilă Jordan

Frontiera lui A este formată

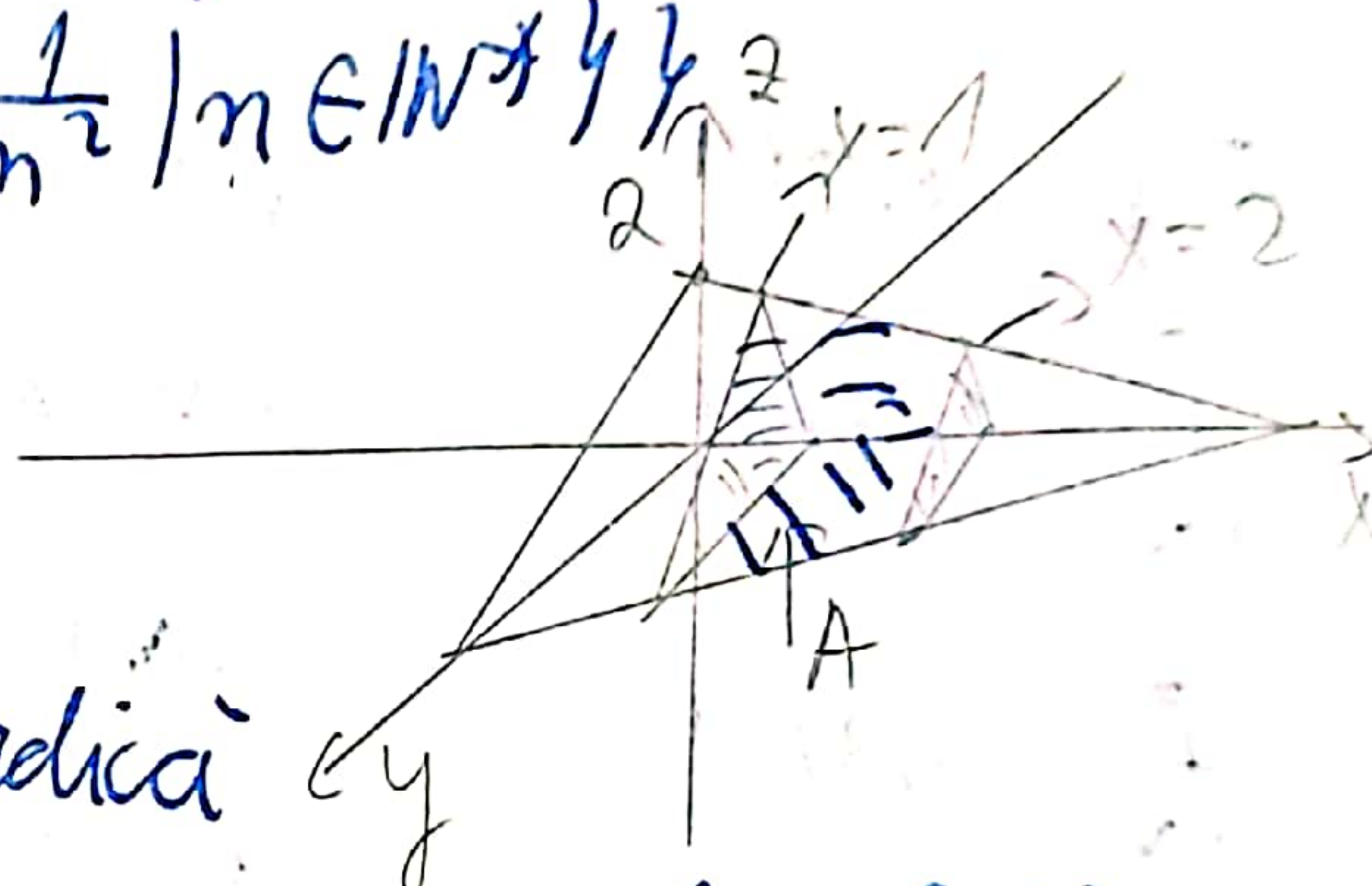
din 2 triunghiuri și 3 patrulatere, adică

din figuri plane. Acestea au măsura Jordan nulă în $\mathbb{R}^3$.

De exemplu, le putem încadra într-un paralelipiped de lățime și lungime suficient de mari (și finite) și de o grosime

neglijabilă $\frac{\epsilon}{L \cdot l} : \epsilon > 0, L \cdot l \cdot \frac{\epsilon}{L \cdot l} = \epsilon \Rightarrow \lambda(\partial A) = 0 \Rightarrow$

$\Rightarrow A$ este măsurabilă Jordan



$$2) \iiint_A x^2 dx dy dz \quad A = \{(x, y, z) \mid x \in [1, 2], y \geq 0, 0 \leq z \leq \frac{8-x-2y}{4}\}$$

$$\Rightarrow I = \iint_{Dxy} \int_0^{\frac{8-x-2y}{4}} x^2 dz dx dy = \iint_{Dxy} x^2 \frac{8-x-2y}{4} dx dy$$

$$= \frac{1}{4} \iint_{Dxy} (8x^2 - x^3 - 2x^2 y) dx dy$$

$$Dxy = \{x, y \mid z=0\} \quad A = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 2y \leq 8, x \in [1, 2]\}$$

$$= \{(x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq \frac{8-x}{2}, x \in [1, 2]\}$$

$$\begin{aligned}
 i &= \frac{1}{4} \int_1^2 \int_0^{\frac{8-x}{2}} (8x^2 - x^3 - 2x^2y) dy dx = \frac{1}{4} \int_1^2 8x^2 \cdot \frac{8-x}{2} - x^3 \cdot \frac{8-x}{2} - x^2 \cdot y^2 \Big|_0^{\frac{8-x}{2}} dx \\
 &= \frac{1}{4} \int_1^2 (32x^2 - 4x^3 - 4x^3 + \frac{x^4}{2} - x^2 (\frac{8-x}{2})^2) dx \\
 &= \frac{1}{4} \int_1^2 (32x^2 - 8x^3 + \frac{x^4}{2} - \frac{x^2}{4} (64 - 16x + x^2)) dx \\
 &= \frac{1}{4} \int_1^2 (32x^2 - 8x^3 + \frac{x^4}{2} - 16x^2 + 4x^4 - \frac{x^6}{4}) dx \\
 &= \int_1^2 (4x^2 - 2x^3 + \frac{x^4}{2} + 4x^4 - \frac{x^6}{16}) dx \\
 &= \left[\frac{4x^3}{3} \right]_1^2 - \left[\frac{2x^4}{4} \right]_1^2 + \left[\frac{3x^5}{10} \right]_1^2 - \left[\frac{x^7}{16 \cdot 7} \right]_1^2 \\
 &= \left(\frac{32}{3} - \frac{4}{3} \right) - \left(8 - \frac{1}{2} \right) + \left(\frac{48}{5} - \frac{3}{10} \right) - \frac{1}{112} \\
 &= \frac{28}{3} - \frac{15}{2} + \frac{93}{10} - \frac{1}{112} \\
 &= \frac{280 - 225 + 279}{30} - \frac{1}{112} \\
 &= \frac{334}{30} - \frac{1}{112} = \frac{167}{15} - \frac{1}{112}
 \end{aligned}$$

3)

IV

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = 5x^2 + 3xy + y^2$$

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\} \quad (\sup_D f(x, y), \inf_D f(x, y))$$

f cont
 D compactă $\} \rightarrow f$ își atinge marginele pe K

$$B((0, 0), 1) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\} \quad \partial B((0, 0), 1) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

I) Căutăm posibile pct. de extrem global ale lui f în $B((0, 0), 1)$.

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 10x + 3y = 0 \quad | \cdot 2 \\ 2y + 3x = 0 \quad | \cdot 3 \end{cases} \Leftrightarrow \begin{cases} 6y + 20x = 0 \\ 6y + 9x = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$\Rightarrow (0, 0)$ este punct critic

$$\frac{\partial^2 f}{\partial x^2} = 10 \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 3 \quad \frac{\partial^2 f}{\partial y^2} = 2$$

$$\Rightarrow H_f(x, y) = \begin{pmatrix} 10 & 3 \\ 3 & 2 \end{pmatrix} \quad \forall (x, y) \in \mathbb{R}^2$$

$$H_f(0, 0) = \begin{pmatrix} 10 & 3 \\ 3 & 2 \end{pmatrix} \Rightarrow \begin{matrix} D_1 = 10 > 0 \\ D_2 = 11 > 0 \end{matrix} \} \Rightarrow (0, 0) \text{ e punct de minim global}$$

$$\Delta: \begin{cases} x = r \cos t & r \in [0, 1] \\ y = r \sin t & t \in [-\frac{\pi}{4}, \frac{\pi}{3}] \end{cases}$$

$$I_3 = \iint_D x^3 dx dy = \int_0^1 \int_{-\pi/4}^{\pi/3} r^3 \cos^3 t \cdot r dt dr$$

$$= \int_0^1 r^4 dr \int_{-\pi/4}^{\pi/3} \cos^3 t dt = \frac{r^5}{5} \Big|_0^1 \int_{-\pi/4}^{\pi/3} \cos t (1 - \sin^2 t) dt$$

$$= \frac{1}{5} \int_{-\pi/4}^{\pi/3} \cos t dt - \frac{1}{5} \int_{-\pi/4}^{\pi/3} \cos t \cdot \sin^2 t dt$$

$$= \frac{1}{5} \sin t \Big|_{-\pi/4}^{\pi/3} - \frac{1}{5} \frac{\sin^3 t}{3} \Big|_{-\pi/4}^{\pi/3} = \frac{1}{5} \left(\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \right) - \frac{1}{5} \left(\frac{(\frac{\sqrt{3}}{2})^3 + (\frac{\sqrt{2}}{2})^3}{3} \right)$$

$$= \frac{9\sqrt{3} + 10\sqrt{2}}{120}$$

$$\Rightarrow I = \iint_{A \cap B} x^3 dx dy = I_1 + I_2 - I_3 = \frac{24}{5} + \frac{24\sqrt{3}}{5} - \frac{9\sqrt{3} + 10\sqrt{2}}{120}$$

$$= \frac{768 + 768\sqrt{3} - 9\sqrt{3} + 10\sqrt{2}}{120} = \frac{759\sqrt{3} + 10\sqrt{2} + 768}{120}$$